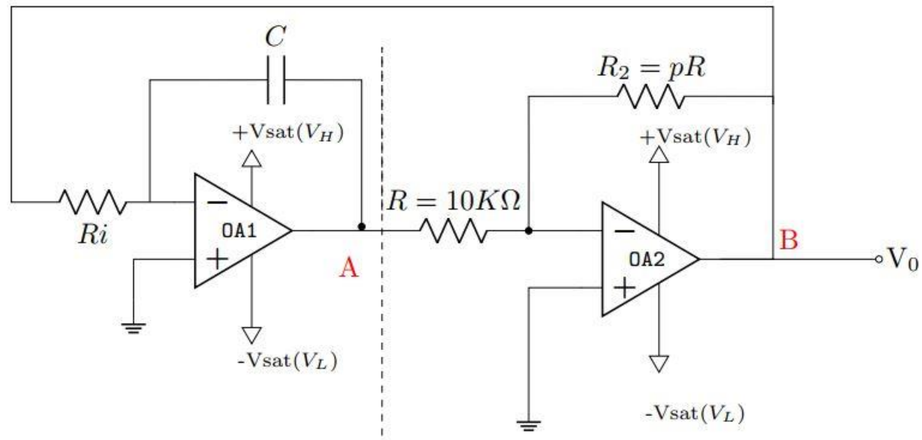


Set-A Question 02

Ri* = 12 KΩ, *R*2 = *pR* = 30 KΩ**, and ***C* = 0.1 *μF

+*Vsat*(*VH*) = +13 *V* and **-*Vsat*(*VL*) = -13 *V***

(a) $V_{UT} = \frac{-R}{R2} * (V_L) = \frac{13}{3} = \boxed{4.333V}$

$V_{LT} = \frac{-R}{R2} * (V_H) = \frac{-13}{3} = \boxed{-4.333V}$

(b) $T_1 = R_i * C * \frac{V_{UT}-V_{LT}}{V_H}$

$= 12k * 0.1u * \frac{4.33+4.33}{13} = \boxed{0.8\ ms}$

$T_2 = R_i * C * \frac{V_{LT}-V_{UT}}{V_L}$

$= 12k * 0.1u * \frac{-4.33-4.33}{-13} = \boxed{0.8\ ms}$

$T = T_1 + T_2 = 1.6\text{ms}$

$F = \frac{1}{T_1+T_2} = \boxed{625\text{ Hz}}$

Duty Cycle $= \frac{T_1}{T_1+T_2} = \boxed{0.5}$

(c) $\boxed{C = \mathbf{0.2\ uF}}$ to make T = 3.2ms (It simply doubles)

Set B Question 01

Ri* = 6 KΩ, *R*2 = *pR* = 40 KΩ**, and ***C* = 0.2 *μF

+*Vsat*(*VH*) = +10 *V* and **-*Vsat*(*VL*) = -10 *V***

(a) $V_{UT} = \frac{-R}{R2} * (V_L) = \frac{10}{4} = \boxed{2.5V}$

$V_{LT} = \frac{-R}{R2} * (V_H) = \frac{-10}{4} = \boxed{-2.5V}$

(b) $T_1 = R_i * C * \frac{V_{UT}-V_{LT}}{V_H}$

$= 6k * 0.2u * \frac{2.5+2.5}{10} = \boxed{0.6\ ms}$

$T_2 = R_i * C * \frac{V_{LT}-V_{UT}}{V_L}$

$= 6k * 0.2u * \frac{-2.5-2.5}{-10} = \boxed{0.6\ ms}$

$T = T_1 + T_2 = 1.2\text{ ms}$

$F = \frac{1}{T_1+T_2} = \boxed{833.33\text{ Hz}}$

Duty Cycle $= \frac{T_1}{T_1+T_2} = \boxed{0.5}$

(c) C = $\boxed{\mathbf{0.1\ uF}}$ to make T = 0.6 ms (It simply halves)

Set A, Question 1

(a) As we neglected base currents of all the transistors,

$$I_{R3} = I_{R4}$$

$$\Rightarrow \frac{0 - V_R}{R_3} = \frac{V_R - 0.7 - V_-}{R_4}$$

$$\Rightarrow \frac{-V_R}{0.7} = \frac{V_R - 0.7 - (-4.1)}{1.3}$$

$$\Rightarrow \boxed{V_R = -1.19\text{ V}}$$

(b) (i) logic HIGH calculation.

at least one of V_x, V_y is HIGH

so, $I_{R_{C2}} = 0$ & Q_R : Cut-off

$$\Rightarrow V_{CR} = 0$$

$$\Rightarrow V_{OR} = 0 - 0.7 = -0.7.$$

$$\therefore \boxed{\text{logic HIGH level} = -0.7}$$

(ii) logic LOW calculation.

$V_x = V_y = \text{LOW}.$

so, $I_{R_{C2}} = I_{RE}$ & Q_R : F.A.

As, $V_R = -1.19V$, $V_E = V_R - 0.7$
$= -1.89\text{ V}$

$$\therefore I_{RE} = \frac{-1.89 - V_-}{R_E}$$

$$= 2.456\text{ mA}.$$

$$\therefore V_{CR} = -I_{RE} \times R_{C2}$$

$$= -1.105\text{ V}$$

$$\therefore V_{OR} = V_{CR} - 0.7 = -1.805\text{ V}$$

$$\boxed{\therefore \text{logic Low level} = -1.805\text{ V}}$$

(c) logic LOW (of V_{NOR}) calculation

atleast one of V_x, V_y is HIGH.

WLOG, suppose $V_x = \text{HIGH} \left(= -0.7\right)$

logic HIGH for both V_{OR}, V_{NOR} is -0.7

$$\Rightarrow V_E = V_x - 0.7 = -1.4\text{V}$$

$$\therefore I_{RE} = \frac{V_E - V_-}{R_E} = 3\text{ mA}.$$

$$\therefore V_{C1} = -I_{RE} \times R_{C1} = -3R_{C1}$$

and $V_{NOR} = V_{C1} - 0.7 = -3R_{C1} - 0.7$

Now, $V_{NOR}(\text{Low}) = V_{OR}(\text{Low}) = -1.805$

$$\Rightarrow -3R_{C1} - 0.7 = -1.805$$

$$\Rightarrow \boxed{R_{C1} = 0.368\text{ k}\Omega}$$

Set B, Question 2

(a) As we neglected base currents of all the transistors,

$$I_{R3} = I_{R4}$$

$$\Rightarrow \frac{0 - V_R}{R_3} = \frac{V_R - 0.7 - V_-}{R_4}$$

$$\Rightarrow \frac{-V_R}{0.5} = \frac{V_R - 0.7 - (-4.5)}{1.5}$$

$$\Rightarrow \boxed{V_R = -0.95V}$$

(b)

(i) logic HIGH calculation.

at least one of V_x, V_y is HIGH

so, $I_{Rc_2} = 0$ & Q_R: Cut-off

$$\Rightarrow V_{CR} = 0$$

$$\Rightarrow V_{OR} = 0 - 0.7 = -0.7.$$

$\therefore$ $\boxed{\text{logic HIGH level} = -0.7}$

(ii) logic LOW calculation.

$V_x = V_y = $ LOW.

so, $I_{Rc_2} = I_{RE}$ & Q_R: F.A.

As, $V_R = -0.95V$, $V_E = V_R - 0.7$
$= -1.65$

$$\therefore I_{RE} = \frac{-1.65 - V_-}{R_E}$$

$$= 3.5625 \text{ mA}.$$

$$\therefore V_{CR} = -I_{RE} \times R_{C2}$$

$$= -2.387 \text{ V}$$

$$\therefore V_{OR} = V_{CR} - 0.7 = -3.087V$$

$\therefore$ $\boxed{\text{logic Low level} = -3.087 \text{ V}}$

(c) logic LOW (of V_{NOR}) calculation

atleast one of V_x, V_y is HIGH.

WLOG, suppose $V_x = $ HIGH $(= -0.7)$

logic HIGH for both V_{OR}, V_{NOR} is -0.7

$$\Rightarrow V_E = V_x - 0.7 = -1.4V$$

$$\therefore I_{RE} = \frac{V_E - V_-}{R_E} = 3.875 \text{ V}$$

$$\therefore V_{C1} = -I_{RE} \times R_{C_1} = -3.875 R_{C_1}$$

and $V_{NOR} = V_{C_1} - 0.7 = -3.875 R_{C_1} - 0.7$

Now, $V_{NOR}(\text{Low}) = V_{OR}(\text{Low}) = -3.087 \text{ V}$

$$\Rightarrow -3.875 R_{C_1} - 0.7 = -3.087$$

$$\Rightarrow \boxed{R_{C_1} = 0.616 \text{ K}\Omega}$$

Question 4

The enhancement-load NMOS compound gate in **figure 4** has 2 input transistors, T_1 and T_2 with inputs , V_1 and V_2 respectively. The transistor parameters are $K_n = 0.2\ mA/V^2$, $V_{TN1} = 0.4V$,$V_{TN2} = 1V$,$V_{TNL} = 0.9V$. Also, The 2-input NMOS compound gate in Figure 4 gives an output of **logical 1**(4 V) only when both inputs are **logical 0** (0 V) **[20]**

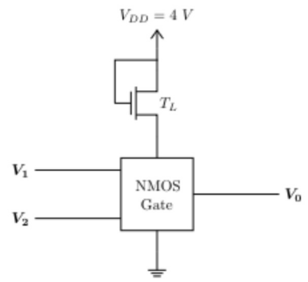


Figure 4

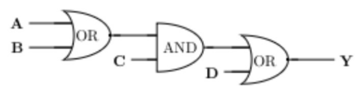


Figure 5

CO2	(a)	**Find** the truth table for the logic gate in figure 4 and derive the expression for output V_0	[2+2]
CO2	(b)	**Find** the operating mode of the load NMOS transistor in figure 4.	[4]
CO2	(c)	In figure 4 **determine** the value of v_0 in V for $V_1 = V_2 = 4\ V$	[4]
CO2	(d)	**Design** an equivalent CMOS logic circuit of Figure 5. (*hint: derive the boolean expression for output Y and use that to design NMOS and PMOS networks for the boolean expression*)	[4+4]

a) Truth table :

V_1	V_2	V_o
0	0	1
0	1	0
1	0	0
1	1	0

2 marks

$V_o = \overline{V_1}\cdot\overline{V_2} = \overline{V_1 + V_2}$ (NOR gate)

2 marks

b) $V_{DS} \geq V_{GS} - V_{TH}$ → $V_{GS_L} = V_{DS_L}$

[saturation]

4 marks

As, $V_{DS_L} > V_{DS_L} - V_{TNL}$; M_L always in saturation.

c) $V_1 = V_2 = 4V$

$I_{DL} = I_{D1} + I_{D2}$

⇒ $k_n (V_{GS_L} - V_{TNL})^2 = k_n\left[2(V_{GS_{D1}} - V_{TN1})V_{DS_{D1}} - V_{DS_{D1}}^2\right] + k_n\left[2(V_{GS_{D2}} - V_{TN2})V_{DS_{D2}} - V_{DS_{D2}}^2\right]$

⇒ $(V_{DD} - V_o - V_{TNL})^2 = [2(4 - 0.4)v_o - v_o^2] + [2(4-1)v_o - v_o^2]$

⇒ $(4 - v_o - 0.9)^2 = 7.2v_o - v_o^2 + 6v_o - v_o^2$

⇒ $(3.1 - v_o)^2 = 13.2v_o - 2v_o^2$

⇒ $9.61 - 6.2v_o + v_o^2 - 13.2v_o + 2v_o^2 = 0$

⇒ $3v_o^2 - 19.4v_o + 9.61 = 0$

⇒ $v_o = 5.92V$ ✗ $> V_{DD}$

① $V_o = \boxed{0.540}$ V

4 marks

SETA

SETB $V_1 = V_2 = 5V$

⇒ $(5 - v_o - 0.9)^2 = 9.2v_o - v_o^2 + 8v_o - v_o^2$

⇒ $(4.1 - v_o)^2 = 17.2v_o - 2v_o^2$

⇒ $16.81 - 8.2v_o + v_o^2 - 17.2v_o + 2v_o^2 = 0$

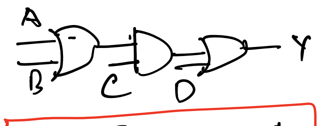
⇒ $3v_o^2 - 25.4v_o + 16.81 = 0$

⇒ $v_o = 7.74V$ ✗ $> V_{DD}$

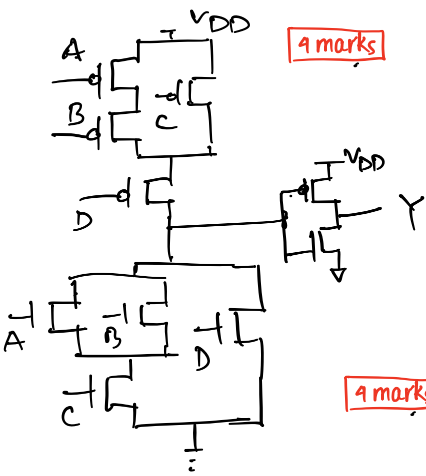
① $V_o = \boxed{0.729}$ V

4 marks

d)

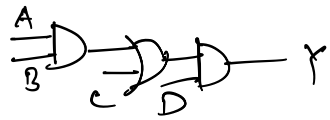


$$\boxed{Y = (A+B)C + D}$$

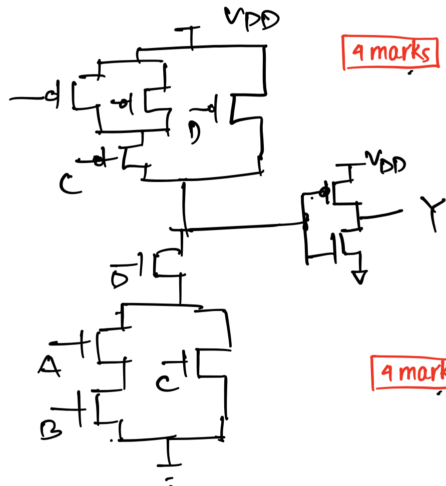


4 marks

4 marks



$$\boxed{Y = (AB + C)D}$$



4 marks

4 marks

Set – A / Question 3

(a) Step size = 8V/7 = 1.1429 V

Quantization Range	Encoding
<1.1429V	000
1.1429V-2.2857V	001
2.2857V-3.4286V	010
3.4286V-4.5714V	011
4.5714V-5.7143V	100
5.7143V-6.8571V	101
6.8571V-8V	110
>8V	111

(b) Here, V_{REF} = 4V

$$V_O = -V_{REF}(b_2 + \frac{b_1}{2} + \frac{b_0}{4})$$

Encoding	**V_O**
000	0V
001	-1V
010	-2V
011	-3V
100	-4V
101	-5V
110	-6V
111	-7V

(c)

V_{IN}	**Encoding**	**V_O**
3.5V	011	-3V
6.7V	101	-5V

Set – B Question 4

(a) Step size = 16V/7 = 2.2857V

Quantization Range	Encoding
<2.2857V	000
2.2857V-4.5714V	001
4.5714V-6.8571V	010
6.8571V-9.1428V	011
9.1428V-11.4285V	100
11.4285V-13.7143V	101
13.7143V-16V	110
>16V	111

(b) Here, V_{REF} = 8V

$$V_O = -V_{REF}(b_2 + \frac{b_1}{2} + \frac{b_0}{4})$$

Encoding	**V_O**
000	0V
001	-2V
010	-4V
011	-6V
100	-8V
101	-10V
110	-12V
111	-14V

(c)

V_{IN}	**Encoding**	**V_O**
5.5V	010	-4V
7.8V	011	-6V